

Subject Description Form

Subject Code	ITC3035T
Subject Title	Materials Foundation II
Credit Value	3 credits
Level	3
Pre-requisite/ <Co-requisite> / (Exclusion)	ITC2021T Materials Foundation I
Objectives	The subject aims to provide students with further knowledge of material science.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - build up a basic knowledge of properties of textile materials - Identify and choose suitable material for textile applications - Communicate effectively with researchers and technologists - Commit to lifelong learning - Develop critical and creative thinking related to material technology
Subject Synopsis/ Indicative Syllabus	(I) Physical properties: - Geometrical optics: reflection and refraction; - Physical optics: Wave function, superposition and interference, diffraction and resolution of optical device, light polarization and application; - Modern optics: Laser, light-material interaction, Raman, luminescence, color and transparent. - Static electric fields: charge, electric field and Coulomb's Law; Electric Potential. - Static magnetic field: the Biot-Savart Law; The Lorentz force equation for a moving charge and applications; - Electric and magnetic fields in materials: Poisson's and Laplace's equation; Capacitance, electrostatic energy and energy density; - Brief introduction to electromagnetic wave; (II) Chemical properties - Step-growth polymerisation: carothers equation, control of molar mass, kinetics, characteristics of step-growth polymerisation. - Free-radical addition polymerisation: addition polymerisation, initiators, chain growth, termination, chain transfer, inhibitors and retarders, heat of polymerisation, features of the free-radical polymerisation. - Ionic polymerisation: general characteristics, cationic polymerisation, anionic polymerisation, living polymers. - Copolymerisation: general characteristics, the copolymer equation, monomer reactivity ratios, reactivity ratios and copolymer structures. - The crystalline state: mechanism of crystallization, melting, crystalline arrangement of polymers, morphology. - The amorphous state: molecular motion, rubbery state, glass transition region, factors affecting T_g, the glassy state.

	7. Structure-property relation: control of melting point (Tm) and glass transition point (Tg), relation between Tm and Tg, block copolymers, plasticizers, crystallinity and mechanical response, application to fibres and elastomers.							
Teaching/Learning Methodology	Lectures, tutorial classes							
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b	c	d	e	
	1. Course work	40%	✓	✓	✓	✓	✓	
	2. Examination	60%	✓	✓			✓	
	Total	100%						
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes: By using appropriately designed course work materials, the knowledge of textiles (outcome a), student's capacity of choosing suitable textiles for application (outcome a), communications (outcome c), the capacity of life-long learning (outcome d) and critical thinking (outcome e) could be assessed. In the final examination, the basic knowledge (outcome a), the application of the learned knowledge for textile material selection (outcome b) and the student's critical and creative thinking (outcome e) could be assessed.							
Student Study Effort Expected	Class contact:							
	▪ Lecture			26 Hrs.				
	▪ Studio			12 Hrs.				
	Other student study effort:							
	▪ Assignments			66 Hrs.				
	Total student study effort			104 Hrs.				
Reading List and References	<u>Books</u> Morton W.E. and Hearle J.W.S., (2008). <i>Physical Properties of Textile Fibres</i> , Fourth Edition, CRC Press. Warner S.B., (1995). <i>Fiber Science, Englewood Cliffs, N.J.</i> , Prentice Hall. Lovell M.C., Avery A.J. and Vernon M.W, (1976). <i>Physical Properties of Materials</i> , Van Nostrand Reinhold Co., Cowie, J.M.G., (2008). <i>Polymers: Chemistry & Physics of Modern materials</i> , 3rd edition, Blackie. Carraher C.E., (2008). <i>Polymer Chemistry</i> , 7th edition, Taylor & Francis							